Phone-Level Mispronunciation Detection for Korean Learners of English

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`https://github.com/lee04jinwoo-sys/aps360-mispronunciation`

Introduction

Learners of English rarely get feedback on *which sound* they got wrong: a teacher can listen to only a few sentences per student, and speaking apps usually return a single sentence-level score. Korean speakers make systematic errors that are easy to name but hard to self-monitor; in the corpus used here the most frequent Korean-L1 substitutions are `DH`→`D` ("*the*"→"*de*", 357 cases), `Z`→`S` (296) and `IH`→`IY` (114). The goal of this project is a model that takes a recording of a read sentence and its expected phone sequence and flags each phone as *correct* or *mispronounced*. Deep learning fits this task because the same phone is realised very differently across speakers, genders and recording conditions; fixed acoustic thresholds such as the classic Goodness of Pronunciation score [5] generalise poorly, whereas a CNN can learn the relevant time–frequency patterns directly from spectrograms.

Illustration

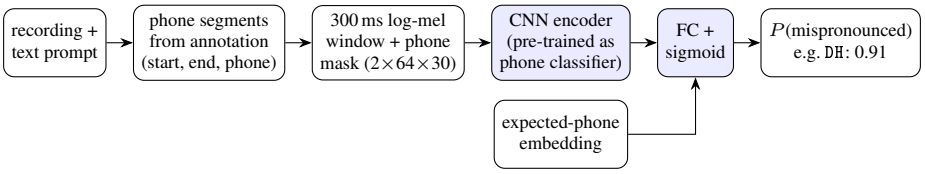


Figure 1: Proposed pipeline. Each expected phone is cut from the utterance with context, converted to a log-mel patch, encoded by a CNN and combined with an embedding of the phone that *should* have been produced.

Background & Related Work

GOP [5] scores a phone by the posterior of the expected phone from an HMM acoustic model and thresholds it; it is still the standard non-neural approach. **Hu et al.** [2] replaced the GMM with a DNN acoustic model and fed GOP-like features into phone-specific logistic-regression classifiers. **Leung et al.** [3] proposed an end-to-end CNN-RNN-CTC phone recogniser whose output is aligned with the expected phones, removing the need for forced alignment. **wav2vec 2.0** [1] learns speech representations from unlabeled audio; **Xu et al.** [6] and **Peng et al.** [4] fine-tuned it for mispronunciation detection and reported strong results on L2-ARCTIC. **L2-ARCTIC** [7] is the non-native speech corpus with phone-level error annotations used in all of these works. In contrast to the large pre-trained models, this project designs a small CNN trained from scratch and studies how much a weakly supervised pre-training step (phone recognition on noisy forced-alignment labels) on *unannotated* utterances of the same corpus helps.

Data Processing

Source. L2-ARCTIC v5.0 [7] (CC BY-NC 4.0): 24 speakers, 6 L1s (Arabic, Mandarin, Hindi, Korean, Spanish, Vietnamese; 2 male + 2 female each), ∼1,130 read sentences per speaker at 44.1 kHz. 3,599

of these utterances have manual phone-level annotations. **Steps.** (1) Parse the `phones` tier of every annotation TextGrid. A label is either a phone (correct) or `CPL,PPL,type` with type `s` (substitution), `d` (deletion) or `a` (addition); stress digits are removed and `err/*` tags kept. This yields 120,243 phone segments: 101,633 correct, 14,098 substitutions, 3,420 deletions and 1,092 additions (15.5% errors; Korean speakers 9.9%). (2) Label each expected phone as mispronounced if it is a substitution or deletion; additions have no expected phone and are excluded. (3) Resample audio to 16 kHz and compute 64-bin log-mel spectrograms (25 ms window, 10 ms hop). (4) Phones are short (median 90 ms, 95th percentile 200 ms), so each sample is a fixed 300 ms window centred on the phone, plus a binary mask channel marking the phone's own frames. (5) Split by speaker so no voice appears in two sets. To compare with prior work [4], the test set is the standard six speakers (NJS, TLV, TNI, TXHC, YKWK, ZHAA; one per L1, 29,786 phones, 13.9% errors), with Korean results reported separately for YKWK; validation = one further speaker per L1; training = the remaining 12 speakers. (6) For pre-training, repurpose the 17,379 *unannotated* utterances of the 18 non-test speakers (test speakers are excluded to avoid leakage): their forced-alignment TextGrids give phone boundaries, so every segment becomes a 40-class phone-recognition example. (7) Handle the 1:5.5 class imbalance with a class-weighted loss; normalise features with training-set statistics.

Architecture

A 2-D CNN encoder takes the $2 \times 64 \times 30$ input (log-mel + mask): four blocks of conv $3\times3 \rightarrow$ batch norm $\rightarrow$ ReLU $\rightarrow$ max-pool (32–128 channels), followed by global average pooling. It is first trained as a 40-way phone classifier on the unannotated non-test utterances. For detection, its 128-d output is concatenated with a learned 16-d embedding of the expected phone and passed through two fully connected layers with dropout and a sigmoid. Training uses Adam and weighted binary cross-entropy, with early stopping on validation F1.

Baseline Model

Per segment, compute 13 MFCCs with first and second derivatives (39 features per frame), take their mean and standard deviation over the phone (78 features) and append a one-hot expected phone (40). A class-weighted logistic regression (scikit-learn) is trained on the same speaker split. Both models are evaluated with precision, recall and F1 on the *mispronounced* class, the standard metrics for this task, overall and per L1, including the Korean test speaker.

Ethical Considerations

The labels encode one annotation team's judgement of "standard" American English; a model trained on them may treat an accent as an error and discourage intelligible but accented speech, so outputs should be presented as feedback, not a grade. With only 24 speakers from 6 L1s, all adults reading the same prompts, performance on other L1s, children, spontaneous speech or other microphones is unknown and likely worse. Voice recordings are biometric data: the corpus licence is non-commercial and forbids redistribution, so the audio is not included in the public repository.

References

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